

Transient Response of First and Second Order AC Circuits	
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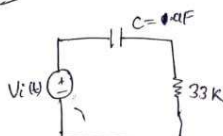
Aim:

Transient Response of First and Second Order AC Circuits

Theoretical Calculation:

Ques 1 Calculate $V_o(t)$ For $t > 0$

$C = 1 \mu F$



frequency = 5 Hz
 $\omega = 2\pi f$
 $\omega = 2\pi \cdot 5$
 $\omega = 10\pi$

$V_m = 1V$
 $V_i(t) = V_m \sin \omega t$
 $V_i(t) = \sin 10\pi t$
 $R = 33k$
 $C = 1 \mu F$

$$V_i(t) = RI(t) + \frac{1}{j\omega C} I(t)$$

$$V_i(t) = \left(R + \frac{1}{j\omega C} \right) I(t)$$

$$Z = R + \frac{1}{j\omega C}$$

$$Z = R + \frac{(-j)}{\omega C}$$

$$|Z| = \sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}$$

$$\phi = \angle - \tan^{-1} \left(\frac{1}{\omega CR} \right)$$

$$I(t) = \frac{V_i(t)}{\left| R + \frac{1}{j\omega C} \right|}$$

$$I(t) = \frac{V_i(t)}{\sqrt{R^2 + \frac{1}{\omega^2 C^2}}} \angle \phi$$

$$I = \frac{1 \angle 0^\circ}{\sqrt{(33 \times 10^3)^2 + \frac{1}{(10\pi)^2 (1 \times 10^{-6})^2}}} \angle \phi$$

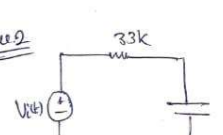
$$I = \frac{1}{\sqrt{2102211836}} \angle \phi$$

$$I = \frac{1 \angle 43.966}{45849.88} = 2.18 \times 10^{-5} A$$

$$I = 21.8 \mu A \angle 43.966$$

Current is leading by $\angle 43.966$

Ques 2



frequency = 5 Hz
 $\omega = 2\pi f$
 $\omega = 10\pi$

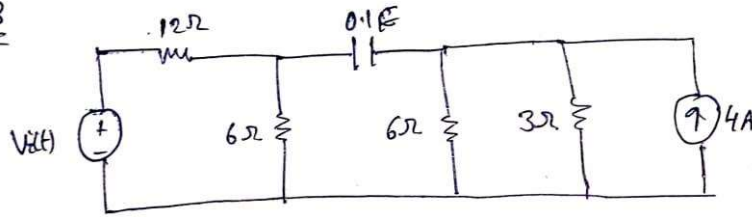
$V_m = 1V$
 $V_i(t) = V_m \sin \omega t$
 $V_i(t) = \sin 10\pi t$
 $R = 33k$
 $C = 1 \mu F$

$$I(t) = \frac{V_i(t) \angle \phi}{\sqrt{R^2 + \frac{1}{\omega^2 C^2}}} \angle \phi$$

$$I = \frac{1 \angle 43.966}{45849.88} = 21.8 \mu A \angle 43.966$$

Current is lagging by phase $\angle 43.966$

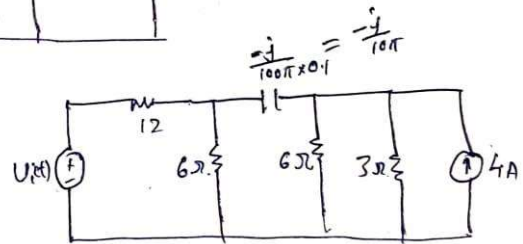
Que 3



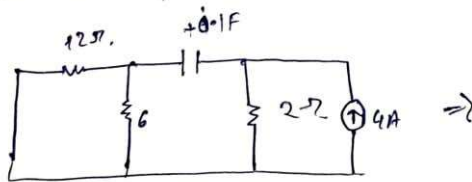
$$\frac{6 \times 3}{8} = 2.25$$

$$V_i(t) = 30 \sin 100\pi t$$

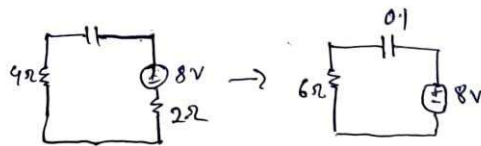
$$\begin{aligned}\omega &= 2\pi f \\ f &= 50 \\ \omega &= 2\pi \cdot 50 \\ \omega &= 100\pi\end{aligned}$$



Due to DC source :-



$$\begin{aligned}V_C(t) &= V(1 - e^{-t/RC}) \\ &= 8(1 - e^{-t/6 \times 0.1}) V\end{aligned}$$



$$\begin{aligned}&= 8(1 - e^{-t/0.6}) V \\ V_C(60ms) &= 0.7613 V\end{aligned}$$

$$\begin{aligned}\frac{1}{6} + \frac{1}{3} &= \frac{3}{6} \\ \frac{12 \times 4}{18} &= \frac{16}{3}\end{aligned}$$

For Due to AC Source.

$$\text{frequency} = 50$$

$$\omega = 2\pi f$$

$$\omega = 2\pi \cdot 50 = 100\pi$$

$$V_i = V_m \sin \omega t$$

$$V_i = 30 \sin 100\pi t$$

$$Z = 12.67 - 9.19j$$

$$I(t) = \frac{V}{Z} = \frac{30 \angle 0^\circ}{12.67 \angle -41.5^\circ}$$

$$= \frac{30 \angle 0^\circ}{15.68 \angle -35.95^\circ}$$

$$I(t) = 1.92 \angle 35.95^\circ$$

$$I_1(t) = \frac{8 \times I(t)}{6 + 2 \frac{-j}{100\pi}} = \frac{6 (1.92 \angle 35.95^\circ)}{8 - \frac{j}{10\pi}} \quad \frac{1}{Z_1} = \frac{6 \times (2 - \frac{j}{100\pi})}{6 + 2 - \frac{j}{200\pi}} + \frac{1}{6}$$

$$I_1(t) = \frac{11.52 \angle 35.95^\circ}{36.83 \angle 77.45^\circ}$$

$$I_1(t) = 0.313 \angle -41.5^\circ$$

$$I_1(t) =$$

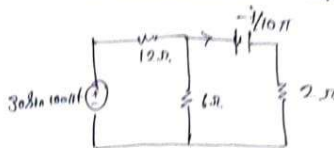
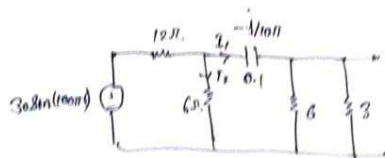
$$V_{C1}(t) = (0.313 \angle -41.5^\circ) \left(-\frac{j}{100\pi} \right)$$

$$V_{C1}(t) = (0.032 \angle 90^\circ) (0.313 \angle -41.5^\circ)$$

$$= 0.01 \angle 48.5^\circ$$

$$V_{C1}(t) = 0.01 \angle 48.5^\circ$$

$$V_{C1}(t) = 6.626 + 7.49j$$



$$Z = \left[\left(2 - \frac{j}{100\pi} \right) \parallel 6 \right] + 12$$

$$Z = \frac{1}{\frac{2 - \frac{j}{100\pi}}{6} + \frac{1}{6}}$$

$$Z_1 = \frac{6 \times (2 - \frac{j}{100\pi})}{6 + 2 - \frac{j}{200\pi}} + \frac{1}{6}$$

$$Z_1 = \frac{8 - \frac{j}{200\pi}}{6(2 - \frac{j}{100\pi})} = \frac{8 - \frac{j}{200\pi}}{12 - \frac{6j}{100\pi}}$$

$$Z_1 = \frac{8 \angle 0.01139^\circ}{12 \angle 0.031188^\circ}$$

$$Z_1 = 0.667 \angle -0.079^\circ$$

$$Z = 0.667$$

$$Z = 0.667 - 9.19j + 12$$

$$Z = 12.67 - 9.19j$$

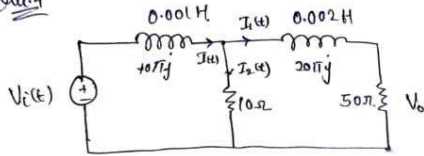
$$V_{C1} = V_{DC} + V_{AC}$$

$$V_{C1} = 0.7613 + 6.626 + 7.49j$$

$$V_{C1}(t) = 7.387 + 7.49j$$

$$V_{C1}(t) = \underline{\underline{8.644 \angle 31.3^\circ}}$$

Ques 4



$$Z = (20\pi j + 50 \parallel 10 \Omega) \text{ series } 10\pi j$$

$$Z = \left(\frac{1}{20\pi j + 50} + \frac{1}{10} \right) + 10\pi j$$

$$Z = \frac{10 + (20\pi j + 50)}{200\pi j + 500} + 10\pi j$$

$$Z = \frac{60 + 20\pi j + (200\pi j + 500)10\pi j}{200\pi j + 500}$$

$$Z = 60 +$$

$$I(t) = \frac{V}{Z} = \frac{10 \angle 0}{31.33 \angle 88.03} = 0.32 \angle -88.03$$

$$I(t) = 0.32 \angle -88.03 \text{ A}$$

Current is lagging by 88.03

$$I_1(t) = \frac{I(t) \times 10}{10 + (20\pi j + 50)}$$

$$I_1(t) = \frac{3.2 \angle -88.03}{60 + (20\pi j)}$$

$$I_1(t) = \frac{3.2 \angle -88.03}{86.87 \angle 46.32}$$

$$I_1(t) = 0.036 \angle -134.35$$

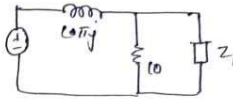
$$V_m = 10 \text{ V} \quad V_i(t) = V_m \sin(\omega t)$$

$$f = 5 \text{ kHz} \quad V_i(t) = 10 \sin(10000\pi t)$$

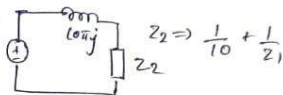
$$\omega = 2\pi f$$

$$\omega = 2\pi \times 5000$$

$$\omega = 10000\pi$$



$$Z_1 = 20\pi j + 50$$



$$Z_2 = \frac{60 + 20\pi j}{20\pi j + 50}$$

$$Z = 10\pi j + Z_2$$

$$Z = 10\pi j + \frac{60 + 20\pi j}{20\pi j + 50}$$

$$Z = 10\pi j + \frac{8687 \angle 46.32}{80.258 \angle 51.43}$$

$$Z = 10\pi j + 1.082 \angle -5.17$$

$$Z = 10\pi j + (1.077 + 0.0975j)$$

$$Z = 1.077 + 31.31j$$

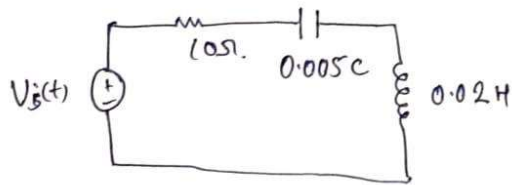
$$Z = 31.33 \angle 88.03$$

$$V_o = I_1(t) \times 50$$

$$V_o = 1.8 \angle -134.35$$

$\Rightarrow V_o$ is lagging by phase 134.35°

Que 5



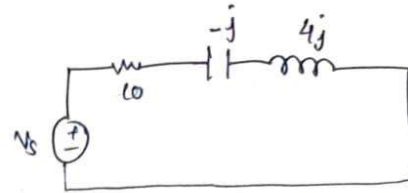
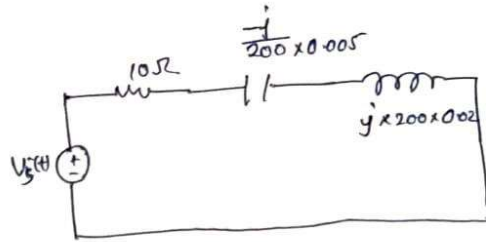
$$\omega = 2\pi f = 200$$

$$V_s(t) = 50 \cos(200t) \\ 50.8 \angle (200t + 90)$$

$$I(t) = \frac{V(t)}{Z} = \frac{50 \angle 90}{10.44 \angle 16.699}$$

$$I(t) = 4.789 \angle 73.301 \text{ A}$$

⇒ Current is leading by $\angle 73.301$.



$$Z = 10 + (j) + 4j$$

$$Z = 10 + 3j$$

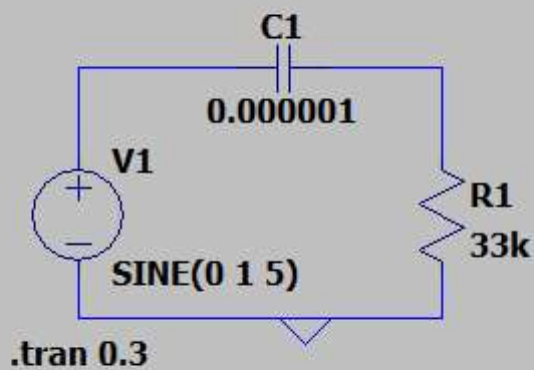
$$Z = 10.44 \angle 16.699$$

Verification by Simulation:

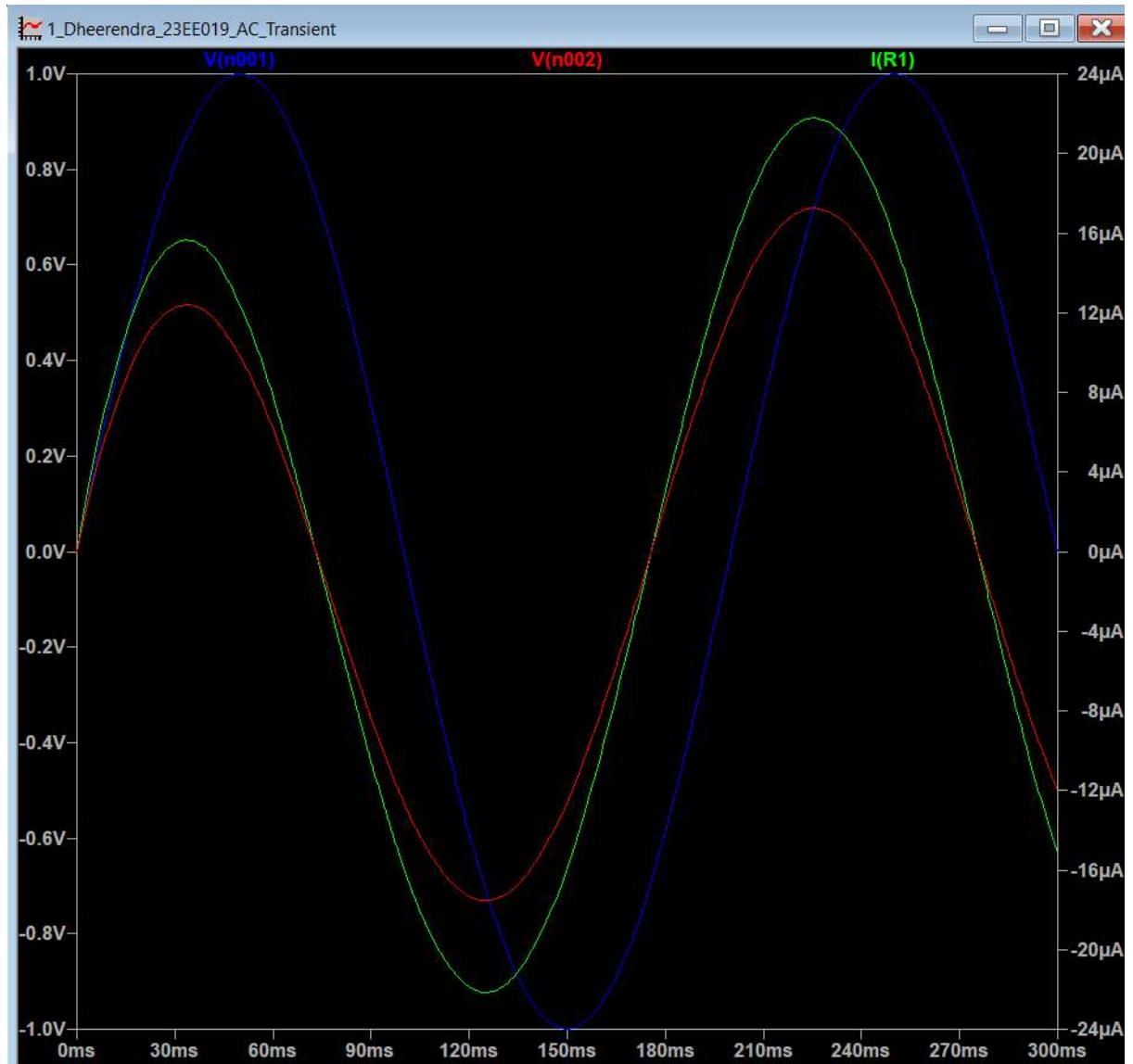
Que 1. Calculate $V_o(t)$ for $t > 0$

Circuit Diagram:

Que 1 calculate $V_o(t)$ for $t > 0$

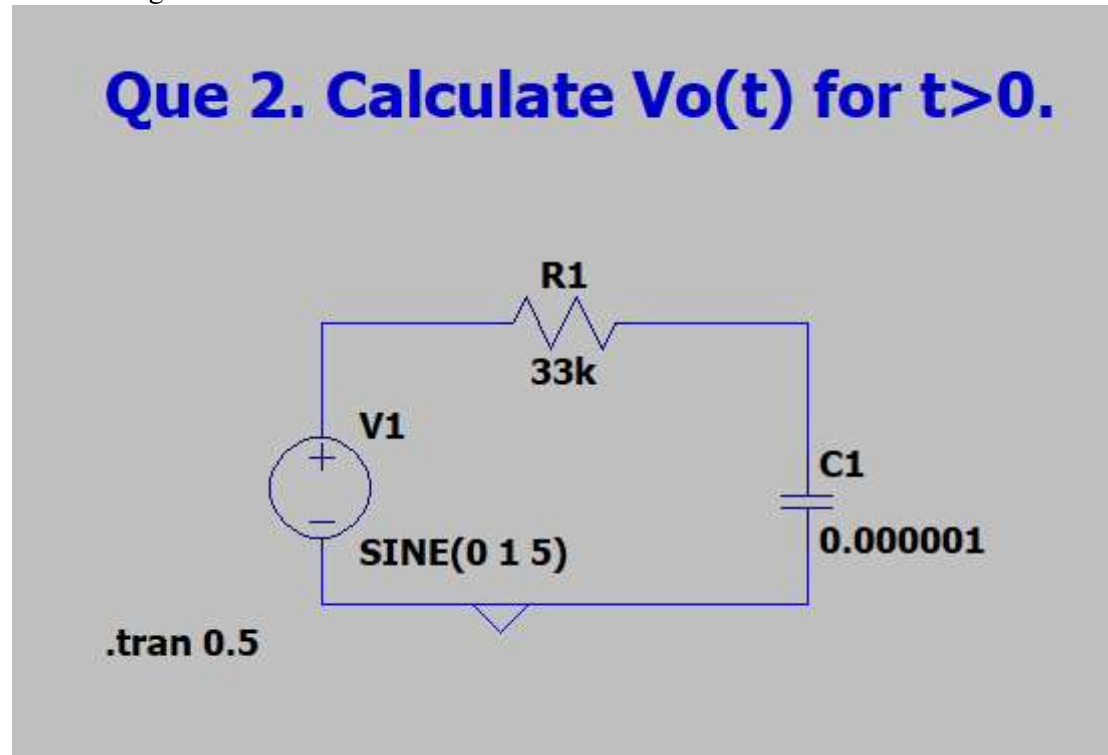


Plot of Voltage/Current:

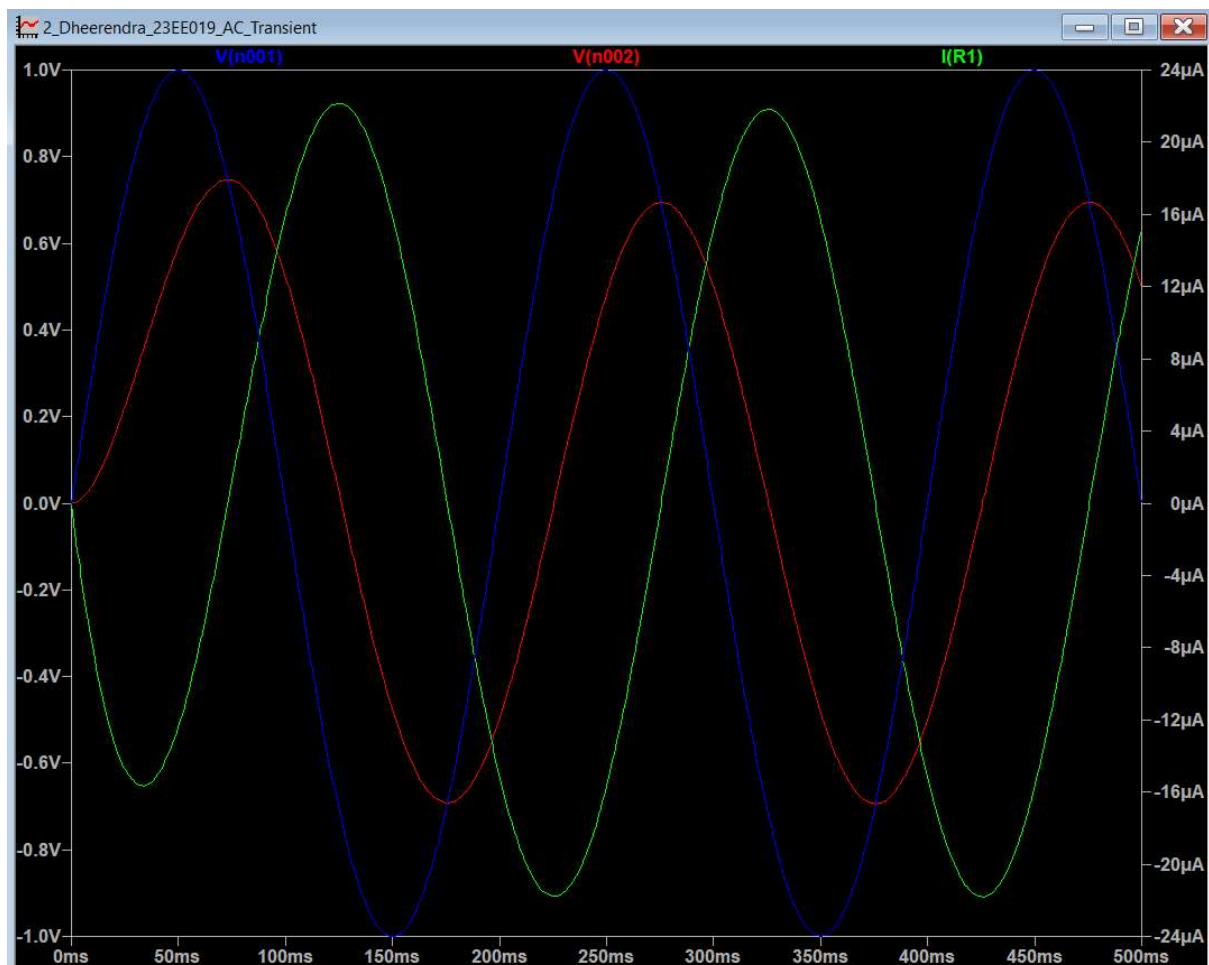


Que 2 Calculate $V_o(t)$ for $t > 0$.

Circuit Diagram:

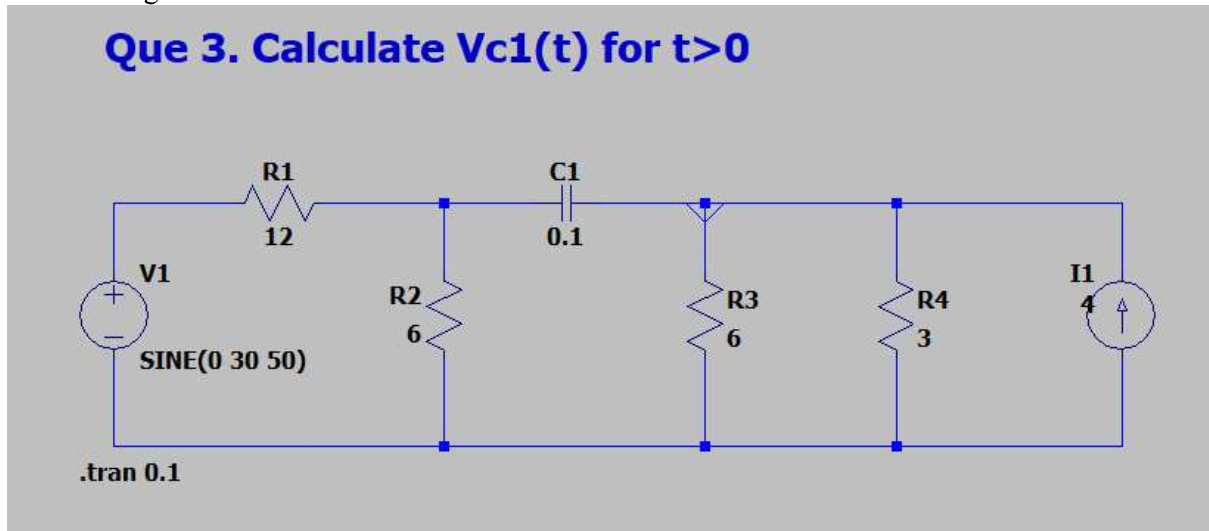


Plot of Voltage/Current:

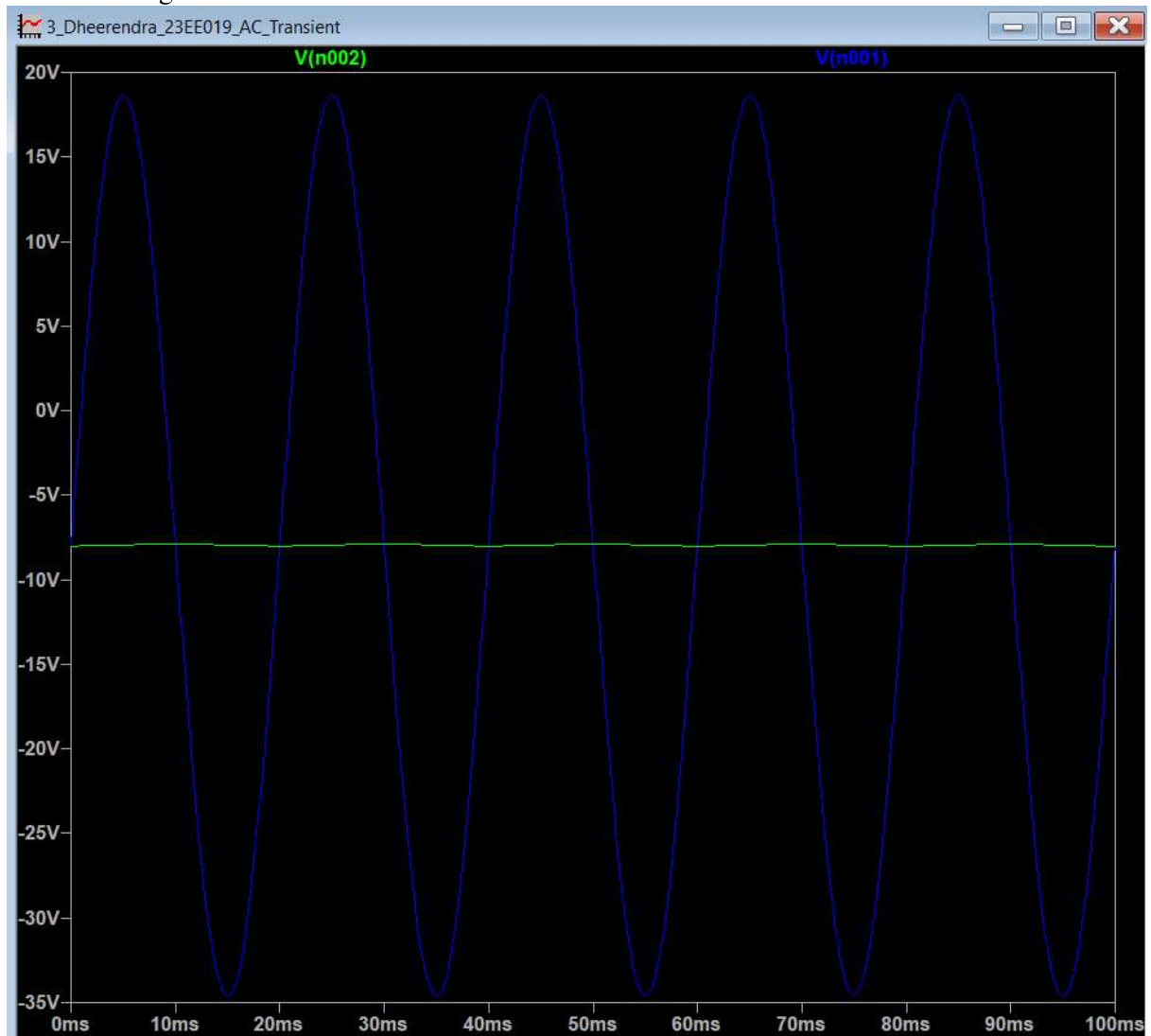


Que 3. Calculate $V_{c1}(t)$ for $t > 0$

Circuit Diagram:

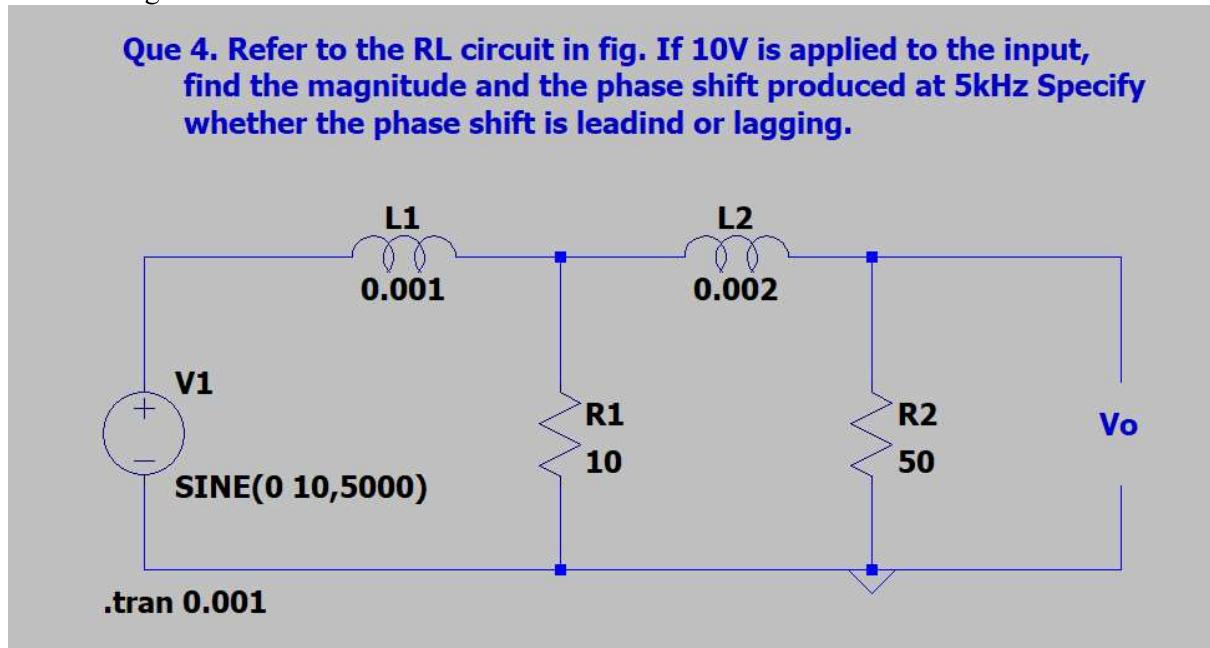


Plot of Voltage/Current:

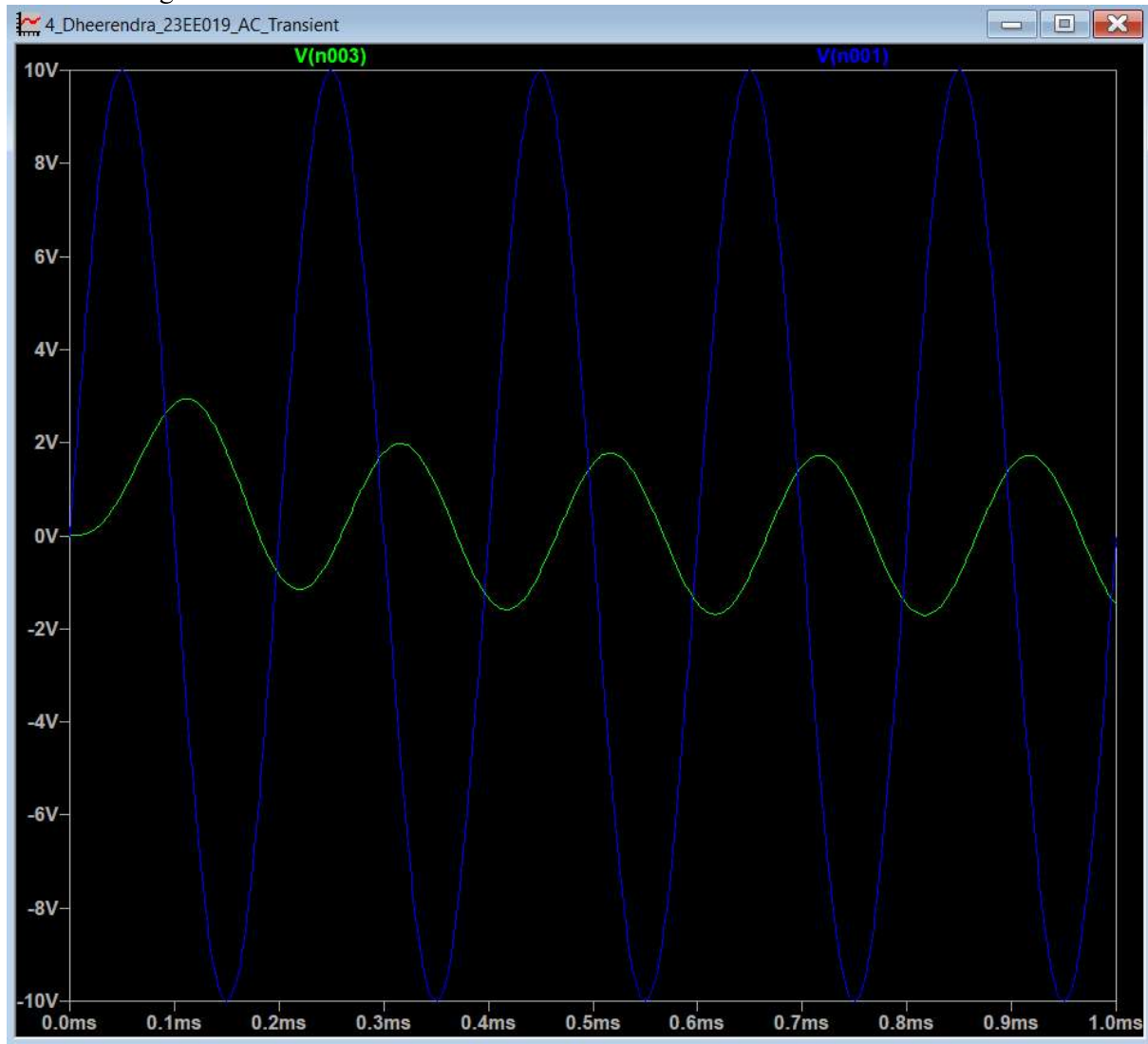


Que 4 Refer to the RL circuit in fig. If 10V is applied to the input, find the magnitude and the phase shift produced at 5kHz Specify whether the phase shift is leading or lagging.

Circuit Diagram:



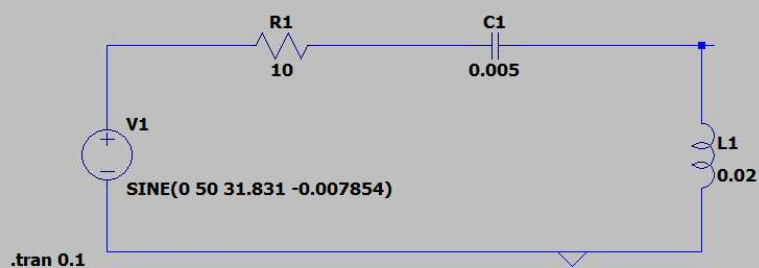
Plot of Voltage/Current:



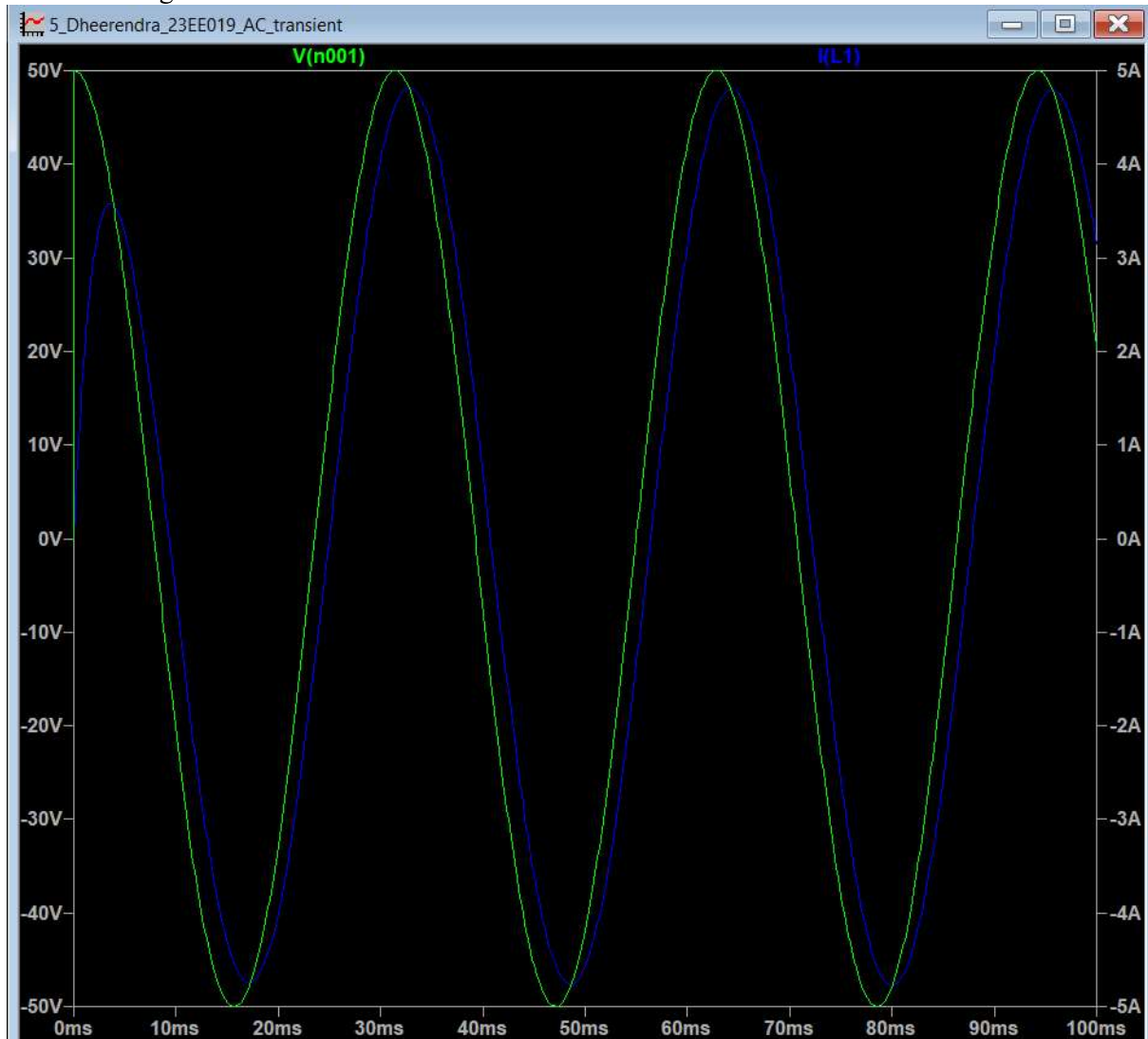
Que 5. find the current i in the circuit of fig. when $V_s(t) = \cos(200t)$ v.

Circuit Diagram:

Que 5. find the current i in the circuit of fig. when $V_s(t) = \cos(200t)$ v.



Plot of Voltage/Current:



Observation Table:

S. No.	$V_o(t)$	$I_L(t)$	Theta (Practical Value)	Theta (Theoretical Value)
1		21.8 μ A	43.2	43.966
2		21.8 μ A	-43.2	-43.966
3	8.644V	-	31.2	31.3
4	1.8V	-	88.5	88.03
5		4.789A	73.1	73.3

Results and Conclusion: As we seen that the theoretical values of phase difference are approximately equal to the practical values of phase difference Hence, we conclude that we have verified the transient response of first order and second order AC circuits.

